Sabrim Mahmud

AI systems developer, multilingual NLP, full-stack development

Dhaka, Bangladesh mahmudsabrim@gmail.com +880 1754 971710 github.com/Sabrimmahmud
linkedin.com/in/sabrim-mahmud-048b88313

Summary

AI systems developer who builds production NLP for Bangla, English and Banglish: ONNX model integration, LLM explanation layers, unified inference APIs, analytics dashboards, and quantized local deployment with Ollama and llama.cpp. First author of two peer-reviewed NLP papers.

Experience

AI Systems Developer Aug 2025 – 2026
Enovosoft, Bangladesh

- **Multilingual sentiment engine.** Developed multilingual sentiment analysis for Bangla, English and Banglish with transformer-based NLP models, served as an ONNX model, with an explanation layer built on the Qwen language model.
- **Unified AI API.** Designed scalable endpoints for text, news and YouTube sentiment analysis, with fallback inference logic.
- **Analytics dashboard.** Built a full-stack sentiment intelligence platform with Next.js, TypeScript, Tailwind CSS and Recharts.
- **Forecasting.** Implemented sentiment trend analysis and time-series forecasting to detect emotional shifts.
- **Model optimization.** Applied bnb-4bit and GGUF quantization and deployed models through Ollama and llama.cpp for efficient inference.

Research Assistant & AI Developer 2024 – 2025
Daffodil International University, AR/VR Lab, Dhaka, Bangladesh

- **AI research.** Worked on hate speech detection, cyberbullying classification and multilingual NLP research.
- **Medical machine learning.** Developed MRI classification pipelines with EfficientNet, YOLO segmentation, and continual learning with EWC and LwF.
- **Full-stack systems.** Built database-driven academic systems and real-time applications.
- **Experimentation and evaluation.** Contributed to AI model experimentation, evaluation and research-based software development.

Projects

UniSentix Multilingual sentiment intelligence suite August 2025 – 2026
A full-stack platform that scores sentiment in Bangla, English and Banglish text, news and YouTube content, explains each prediction with a language model, and tracks how sentiment shifts over time.
ONNX, Qwen, Ollama, llama.cpp, bnb-4bit, GGUF, Next.js, TypeScript, Tailwind CSS, Recharts

YouTube Transcript Analyzer Segment-level sentiment and emotion timelines for video
Scores sentiment for each segment of a YouTube transcript and plots the results as an emotional timeline, so you can see where a video's tone changes.

Medical AI Pipeline MRI classification, segmentation and continual learning 2024 – 2025
A pipeline for MRI image classification and feature extraction that tests whether a model can learn new tasks without forgetting earlier ones.
EfficientNet, YOLO, EWC, LwF

Technical skills

Programming	Python, TypeScript, JavaScript, C++, Java, SQL
Artificial intelligence	PyTorch, Transformers and fine-tuning, Natural language processing, ONNX, Quantization (bnb-4bit, GGUF), LLM integration (Qwen), Time-series forecasting, EfficientNet and YOLO, Continual learning (EWC, LwF)
Frontend	React, Next.js, Tailwind CSS, Recharts
Backend	REST API design, FastAPI, Node.js
Tools and deployment	Ollama, llama.cpp, Git, MySQL

Publications

PoliticaNet: A Transformer-Driven Framework for Fine-Grained Classification of Bangla Political Hate Speech in Social Media. First author. APANConf 2025, Springer, 2026. DOI: 10.1007/978-3-032-18319-4_6

Identifying Bangla Political Hate Speech using Transformer-based Approach. First author. NCIM 2025, IEEE, 2025. DOI: 10.1109/NCIM65934.2025.11159848

Education

Bachelor of Science in Computer Science and Engineering Jan 2022 – Jan 2026
Daffodil International University, Dhaka, Bangladesh
CGPA 3.84 / 4.00, 148 of 148 credits completed
Undergraduate thesis, industrial R&D project: *UniSentix: A Multilingual and Multimodal Sentiment Analysis Framework with Explainable Insights*

Higher Secondary Certificate (HSC) 2020
Rajshahi Govt. City College, Rajshahi, Bangladesh
Science group, GPA 5.00 / 5.00

Secondary School Certificate (SSC) 2017
Akkelpur F. U. Pilot High School
Science group, GPA 5.00 / 5.00